MGU-UGP (HONOURS) 3rd Semester Regular EXAMINATION October 2025
Value Addition Course (VAC) - MG3VACCSC200 - WHITE HAT HACKING
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Scheme of Evaluvation

Part A

Very Short Answer Questions, Answer all questions, Each question carries 2 marks

1. Define Target of Evaluation (TOE). [K/CO1]

Expected: "System, program, or network under security assessment/attack focus."

Partial 1 mark: Vague "target" without security-analysis context

CEH Sybex: Ch.1, Page No.7.

2. Name one piece of information collected during footprinting. [K/CO2]

Expected: Any one of these "domain name, IP range/network blocks, DNS records, OS hints, contact names, emails, phone numbers, exposed services, web server type."

Partial 1 mark: Generic "information" without specificity

CEH Sybex: Ch.2, Page No.38.

3. Differentiate between wired and wireless eavesdropping. [U/CO3]

Core: Wired requires access to switch/tap/SPAN; wireless can be captured over-the-air; wireless adds RF range/SSID/MAC issues; encryption differences (WEP/WPA).

Marking:

Full 2: States medium difference and access modality (cable/switch vs RF), mentions implication (range/encryption)

Partial 1 mark: Mentions only "wireless is over air" without contrast

CEH Sybex: Ch.6 pp.154–166; Ch.10 overview for wireless.

4. Define the term cracker. [K/CO1]

Expected: Malicious hacker using skills for illegal/destructive purposes.

Partial 1 mark: "Hacker" without malicious element

CEH Sybex: Ch.1, Page No.3.

5.List the different types of scanning. [K/CO2]

Expected: Network scanning, Host scanning & Port scanning

Partial 1 mark: Any common scanning techniques

CEH Sybex: Ch.3, Page No.64.

Part B

Short Answer Questions, Answer any 5 questions, Each question carries 5 marks

Awarding bands per question:

- ***5 Marks: Accurate, structured 3–5 key points, correct examples/tools,***
- ***3–4 Marks : Mostly correct 2–3 points, minor gaps***
- ***1–2 Marks: Fragmented, 1–2 points, limited relevance***

- *0 Mark: Off-topic/incorrect*

6. Discuss how a hacker can use an attack to exploit a known vulnerability. [U/CO1]

Expected points: vulnerability→exploit→payload; patch state; CVE mapping; delivery vectors (remote/local); outcome (CIA impact). Example (e.g., SMB vuln → remote code exec).

Minimum elements for Full Marks: define vulnerability/exploit, vector example, impact + countermeasure (patching).

CEH Sybex: Ch.1 pp.7–12; Ch.4 overview.

7. Describe Information Gathering Methodology. [U/CO2]

Seven steps outline: unearth initial info → locate network range → ascertain active hosts → discover open ports → detect OS → uncover services → map network. Include passive vs active footprinting.

Minimum elements for Full Marks: clear sequence with brief purpose of each step.

CEH Sybex: Ch.2 pp.37–48; Fig.2.1.

8. Interpret how redirection attacks compromise network communication. [U/CO3]

Concepts: traffic redirection via ARP spoofing/SMB relay/DNS spoofing/HTTP 302 abuse; man-in-the-middle; session hijack implications; confidentiality/integrity impact.

Minimum elements for Full Marks: define redirection, one L2 (ARP) and one L7/L3 example, impact + countermeasure (ARP inspection/DNSSEC/HTTPS+HSTS).

CEH Sybex: Ch.6 (ARP/DNS), Ch.7 (session hijacking).

9. Explain why information theft is a confidentiality attack, with examples. [U/CO1]

Concepts: Map it to CIA triad; examples: sniffed passwords, database exfiltration, stolen backups, shoulder surfing; contrast with integrity/availability.

Minimum elements for Full Marks: define confidentiality, 2–3 examples, brief mitigation (encryption, access control).

CEH Sybex: Ch.1 pp.5–7.

10. Describe the performance of reconnaissance tools in white hat hacking. [U/CO2]

Tools and roles: Whois, DNS lookup/NSlookup, traceroute, search engines/Google dorks, Netcraft, EDGAR, OSINT; outputs and how they guide scanning.

Minimum elements for Full Marks: 3–4 tools with what they reveal and how they feed next steps.

CEH Sybex: Ch.2 pp.38–48; Ch.3 linkage.

11. Differentiate brute force and dictionary attacks with examples. [U/CO3]

Contents: Definition, process, complexity, time, hybrid/salted hashes impact; examples/tools (Hydra, John, Hashcat); mitigation (account lockout, strong passwords).

Minimum elements for Full Marks: define both, contrast, one example each, mitigation.

CEH Sybex: Ch.4 pp.96–104, 107

12. Detecting live hosts using ping sweeps and list any two commonly used tools. [U/CO2]

Concept: ICMP Echo Request across range; note firewall/ICMP block; alternatives (ARP scan, TCP SYN to common ports); tools: Nmap, fping/hping/Netdiscover.
Minimum elements for Full Marks: method description, limitations, 2 tools named, brief command example earns full

CEH Sybex: Ch.3 pp.68–70..

Part C

Essay Questions, Answer any 1 questions, Each question carries 15 marks

Awarding bands:

- **13–15: Comprehensive structure, accurate concepts, 5–7 well-developed points, correct examples/tools/mitigations, conclusion**
- **9–12: Good coverage, 3–5 points, minor gaps**
- **5–8: Basic coverage, few points, limited examples**
- **0–4: Minimal/incorrect**

13. Explain the importance of scripting in ethical hacking with examples. [U/CO2]

Scope: automation of reconnaissance/scanning; parsing results; quick PoCs; reporting; safe task repetition; examples in Bash/Python/PowerShell (e.g., mass Nmap wrapper, log parser, API calls); risks/ethics.
Suggested outline for full marks:

- Why scripting matters (speed, repeatability, accuracy)
- Where used (recon, enumeration, exploitation staging, evidence collection)
- 2–3 concrete examples
- Safety/ethics and documentation

14. Differentiate DNS, IP, and HTTPS sniffing; illustrate attack scenarios. [U/CO3]

DNS sniffing: capture queries/responses; risks (hostnames, subdomains), spoofing/poisoning angle.
IP sniffing: passive capture at L3/L2; unencrypted protocols; ARP poisoning to get in-path; session hijack risk.
HTTPS sniffing: normally protected; scenarios where visible (misconfig, SSL stripping, rogue CA, compromised proxy); HSTS role.
Tools: Wireshark, tcpdump, mitmproxy, Ettercap.
Mitigations: encryption, DNSSEC/DoT/DoH, ARP inspection, switch port security, HSTS, certificate pinning.
For 13–15: clear distinctions, one scenario each, tool + countermeasure per type.

CEH Sybex: Ch.6 pp.158–166 (sniffing, ARP, DNS), Ch.7 hijacking.